

1. Prove that $2^{n+2} + 3^{2n+1}$ is divisible by 7 for every $n \geq 0$.

2. Consider the Fibonacci sequence $(1, 1, 2, 3, 5, 8, 13, \dots)$; that is, let $(f_1, f_2, f_3, \dots)$ be the sequence defined by $f_1 = 1$, $f_2 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 3$.
 1. Prove by induction that every fifth element of the sequence is divisible by five; i.e. prove that $5 \mid f_{5n}$ for each $n \geq 1$.
 2. Prove by induction that $f_1 + f_2 + f_3 + \dots + f_n = f_{n+2} - 1$ for all $n \geq 1$.
 3. Prove by induction that $f_1^2 + f_2^2 + \dots + f_n^2 = f_n f_{n+1}$ for all $n \geq 1$.

3. Prove by induction that, for all $n \in \mathbb{N}$, we have $(1 + 2 + 3 + \dots + n)^2 = 1^3 + 2^3 + 3^3 + \dots + n^3$.